Aldehydes, ketones & Carboxylic Acid

MOP of Aldehydes & Ketones

① By oxidation of Alcohols

1° OH $R-\boxed{CH_2-OH} \xrightarrow{M.O.A} R-\boxed{CHO}$

2° OH $R-\boxed{\underset{|}{CH}(OH)}-R' \xrightarrow[\text{S.O.A}]{\text{M.O.A or}} R-\boxed{\overset{}{C}(=O)}-R'$

3° OH $R-\overset{OH}{\underset{R''}{C}}-R' \longrightarrow$ NO Rxn

② By dehydrogenation of Alcohols

Reagent:- Cu or Ag / 300°C

Removal of $H_2 \rightarrow$ oxidation

1° OH $\longrightarrow R-CH_2-OH \longrightarrow R-CHO$ Aldehyde

2° OH $\longrightarrow R-\underset{OH}{CH}-R' \longrightarrow R-\underset{O}{\overset{}{C}}-R'$ ketone

3° OH $\longrightarrow$ >—OH $\longrightarrow$ >= Alkene

③ From hydrocarbons

ⓐ Ozonolysis (Reductive)

Reagent:- $O_3 + CCl_4$ / Zn or Zn + H_2O or Me_2S

$>C=C< \longrightarrow >=O + O=<$

ⓑ Kucherov Reaction

↳ Hydration of Alkyne

$R-C\equiv CH \xrightarrow[\text{dil. } HgSO_4]{\text{dil. } H_2SO_4} R-\overset{OH}{C}=CH_2 \rightleftharpoons R-\overset{O}{\overset{||}{C}}-CH_3$

H^+ OH^-

(1 time)

MOP of Aldehydes

① Rosenmund Reduction

$R-\overset{O}{\overset{||}{C}}-Cl \xrightarrow{H_2 + Pd/BaSO_4} R-\overset{O}{\overset{||}{C}}-H$

[Lindlar's Catalyst]

Cl hatao H lagao

② Stephens Reduction

$R-C\equiv N \xrightarrow[H_3O^+ \text{ or steam}]{SnCl_2 + HCl} R-CHO$

CN hatao, CHO lagao

③ DIBAL-H :- Alternate of Stephen's reduc

↳ Diisobutyl aluminium hydride

$Al(ibu)_2-H$ Reagent:- DIBAL-H / H_2O

ⓐ Rxn with nitrile

ⓑ Rxn with Ester

$R-\overset{O}{\overset{||}{C}}-OR' \longrightarrow R-\overset{O}{\overset{||}{C}}-H + R'OH$

Ester $\xrightarrow{DIBAL-H}$ Aldehyde

④ From Hydrocarbons

① Etard Reaction

$Ph-CH_3 \xrightarrow[H_3O^+]{CrO_2Cl_2 / CS_2} Ph-CHO$

Intermediate:- Chromium complex

$PhCH[OCr(OH)Cl_2]_2$

OP Point:-

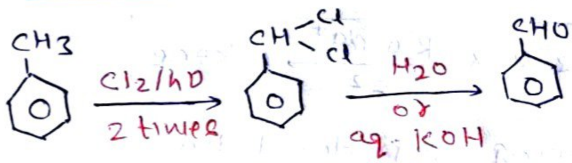
Alternate reagent:- $CrO_3 + (CH_3CO)_2O$

Intermediate:- Benzylidene diacetate

$PhCH(OCOCH_3)_2$

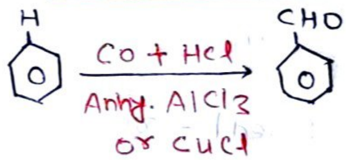
CH_3 Hatao, CHO lagao

② Side chain oxidation



CH_3 Hatao, CHO lagao

③ Gatterman Koch Reaction



H hatao CHO lagao

MOP of Ketones

① From acyl chloride

$2RMgCl \xrightarrow{CdCl_2} R_2Cd + 2MgCl_2$

$2\,R'-\overset{O}{\overset{||}{C}}-Cl + Cd\begin{matrix}R\\R\end{matrix} \longrightarrow 2R'-\overset{O}{\overset{||}{C}}-R + CdCl_2$

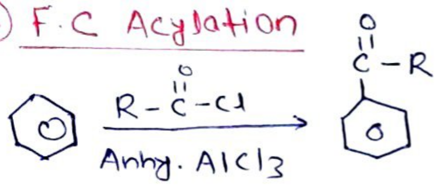
Acid chloride ka 'cl' hatao, and G.R ka R_2Cd ka R lagao

② G.R (Reaction with cyanide)

$$R'-C\equiv N \xrightarrow[H_3O^+]{RMgX} R'-\underset{R}{\underset{|}{C}}=O$$

G.R ka R, CN ke C se jodkar ketone bna denge

③ F.C Acylation



H Hatao COR lagao

Properties of Aldehydes & ketones

① NAR

$$>C=O \xrightarrow{Nu^\ominus} >\underset{Nu}{\underset{|}{C}}-O^\ominus \xrightarrow{H^+} >\overset{Nu}{C}-OH$$

sp^2 — Alkoxide intermediate — sp^3

Reactivity :-

(i) In NAR → (Ald > ket)

(ii) ROR $\propto \frac{1}{SH}$

(iii) ROR $\propto$ Electrophilicity $\propto -M \propto -I$

ROR → HCHO > CH_3CHO > CH_3COCH_3

★ Some important examples of NAR

① Addition of hydroge cyanide (HCN) :-

Reagent :- (i) HCN (ii) NaCN or KCN / H_2O

$$>\overset{+}{C}=\overset{-}{O} \xrightarrow{\overset{+}{H}\overset{-}{CN}} >\underset{CN}{\underset{|}{C}}-OH \text{ (cyanohydrin)}$$

★ op point :-

$$-CN \xrightarrow{\text{Partial hydro.}} -\overset{O}{\overset{||}{C}}-NH_2$$

$$-CN \xrightarrow{H_3O^+} -COOH$$

② Addition of $NaHSO_3$:-

↳ $NaHSO_3$ test given :-

(i) All Aldehyde

(ii) Methyl ketone

$$>\overset{+}{C}=\overset{-}{O} \xrightarrow{H^+\ \overset{-}{SO_3}Na} >\underset{SO_3Na}{\underset{|}{C}}-OH$$

③ Addition of alcohols

$$>\overset{+}{C}=\overset{-}{O} \xrightarrow[H^+]{ROH} >C\begin{matrix}-OH\\-OR\end{matrix}$$ Hemi acetal or Hemi ketal

$$>C=O \quad \begin{matrix}HOR'\\HOR'\end{matrix} \longrightarrow >C\begin{matrix}-OR'\\-OR'\end{matrix}$$ Acetal or ketal

⇒ Hydrolysis of Hemi acetal, Hemi ketal, acetal, ketal :-

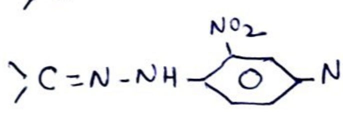
↳ All will become carbonyl again.

$$\begin{matrix}R\\H\end{matrix}>C\begin{matrix}OH\\OR\end{matrix} \xrightarrow{H_3O^+} \begin{matrix}R\\H\end{matrix}>C=O$$

Hemi-acetal — Carbonyl

④ Addition of ammonia & its derivatives

$$>C=O \xrightarrow{H_2N-Z} >C=N-Z$$

Z (derivatives)	product name
$>C=NH$ →	Imine
$>C=NR$ →	Schiff's base
$>C=N-OH$ →	Oxime
$>C=N-NH_2$ →	Hydrazone
$>C=N-NH-Ph$ →	Phenylhydrazone
 →	2,4 Dinitrophenyl hydrazone
$>C=N-NH-\overset{O}{\overset{||}{C}}-NH_2$ →	Semicarbazone

② Reduction Reactions

ⓐ Reduction by LAH, SBH & H_2 + Catalyst

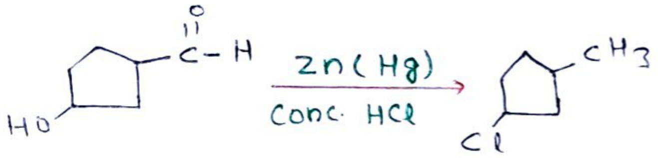
$$>C=O \longrightarrow >\underset{H}{\underset{|}{C}}-OH$$

CO hatao CHOH lagao

ⓑ Clemmenson's Rxn

$$>C=O \xrightarrow[\text{Conc. HCl}]{Zn(Hg)} >CH_2$$

CO hatao CH_2 lagao

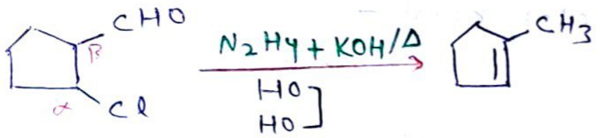


② Wolff-Kishner Reduction

Reagent:- NH_2NH_2 / KOH, HO HO]

$>C=O \longrightarrow >CH_2$

* OP Point:-



③ Oxidation Reaction

① Tollen's Test

Reagent:- Tollen's reagent

Ammoniacal silver nitrate

$(AgNO_3 + NH_4OH)$ (O.A)

Silver mirror test

**
Ali Ald ✓
Aro Ald ✓
ketone ✗

$-\overset{O}{\overset{\|}{C}}-H \xrightarrow{Ag^+} -\overset{O}{\overset{\|}{C}}-O^{\ominus} + \overset{0}{Ag}$

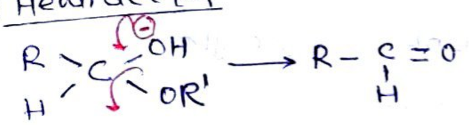
Silvery white

Redn

oxi

Other comp. perform this test

ⓐ Hemiacetal



ⓑ α-hydroxy ketone

$R-\underset{\alpha}{\overset{OH}{CH}}-\overset{O}{\overset{\|}{C}}-R' \longrightarrow R-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-R'$

ⓑ Fehling's Test

Fehling A + Fehling B

aq $CuSO_4$ NaKT

Ali Ald ✓
Aro Ald ✗
ketone ✗

$-CHO \rightarrow -COO^{\ominus}$ [Rochelle's salt]

$\overset{2+}{Cu} \rightarrow \overset{+1}{Cu_2O}$

Red Brown ppt.

COONa
CHOH
CHOH
COOK

Hemiacetal ✓
α-hydroxy ket ✓

$R-CHO \xrightarrow[\text{fehling B}]{\text{fehling A}} Cu_2O + RCOO^{\ominus}$

ⓒ Haloform Test

Reagent:- $X_2 + NaOH/KOH$ or $CaOCl_2$

or

$NaOX / KOX$

↳ +ve Haloform Test is given by

① Methyl ketones → $R-\boxed{\overset{O}{\overset{\|}{C}}-CH_3}$

② 2° Alcohols → $R-\boxed{\underset{OH}{CH}-CH_3}$

X_2 (M.O.A)

③ 1° Alcohol Ethanol → $H-\boxed{\underset{OH}{CH}-CH_3}$

④ Acetaldehyde → $H-\boxed{\overset{O}{\overset{\|}{C}}-CH_3}$

X_2 (M.O.A)

★ OP Points:-

① Carbanion intermediate formation

② R.O.R → $CHF_3 \approx CHCl_3 \approx CHBr_3 \approx CHI_3$

③ Net Rxn

$R-\overset{O}{\overset{\|}{C}}-CH_3 + 3X_2 + 4NaOH \longrightarrow RCOONa + CHX_3 + 3NaX$

D.M

① Halogenation

$R-\overset{O}{\overset{\|}{C}}-C\boxed{\begin{matrix}H & X-X\\ H & X-X\\ H & X-X\end{matrix}} \longrightarrow R-\overset{O}{\overset{\|}{C}}-CX_3$

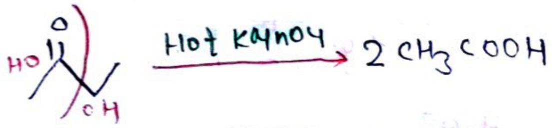
H hatao
X lagao

② Alkaline hydrolysis

$R-\underset{ONa}{\overset{O}{\overset{\|}{C}}}+\underset{H}{CX_3} \longrightarrow RCOONa + CHX_3$

④ Popoff's Rule

Reagent:- [Hot $KMnO_4$ or Acidic $KMnO_4$]



⑤ Strong oxidizing agents

Ald ⟶ Acid

$R-CHO \xrightarrow{S.O.A} R-COOH$

ALDOL CONDENSATION

① Aldol condensation

↳ carbonyl comp. having α-H

↳ Reagent :- dil. Base ($NaOH, KOH, Ba(OH)_2$) etc.

$$2CH_3-\overset{O}{\overset{\|}{C}}-H \xrightarrow{dil.\,NaOH} (A) \xrightarrow{\Delta} (B)$$

(A) $\overset{\beta}{C}H_3-\overset{\beta}{C}H(OH)-\overset{\alpha}{C}H_2-CHO$ — β-hydroxy carbonyl comp.

(B) $CH_3-\overset{\beta}{C}H=\overset{\alpha}{C}H-CHO$ — α, β - unsaturated carbonyl comp.

* OP Point :-

① Carbanion formation (R.D.S)

② Aldol product is stable due to intramolecular H-bonding.

D.M

① $>C=O + H-Z \rightarrow >C(Z)-O-H \xrightarrow{\Delta}$ Alkene

$$CH_3(H)C=O + H-CH_2CHO \xrightarrow[NaOH]{dil.} CH_3(H)C(OH)CH_2CHO \xrightarrow{\Delta} CH_3-CH=CHCHO$$

② $>C=\boxed{O\ \ H_2}Z \xrightarrow[\Delta]{dil.\,NaOH} >C=Z$

② Cross aldol condensation

$2CH_3CHO + 2CH_3CH_2CHO \xrightarrow[\Delta]{dil.\,NaOH}$ total no. of products

① cross product → 2
② self product → 2] → ④

$CH_3(H)C=O + H_2C(Me)-CHO \rightarrow CH_3-CH=C(Me)-CHO$ (cross)

$CH_3(H)C=O + H_2CH-CHO \rightarrow CH_3-CH=CH-CHO$ (self)

$Et(H)C=O + H_2C(Me)-CHO \rightarrow Et-CH=C(Me)-CHO$ (self)

$Et(H)C=O + H_2CH-CHO \rightarrow Et-CH=CH-CHO$ (cross)

③ Cannizzaro Reaction

↳ Aldehydes do not have α-H

Reagent :- conc. $KOH/NaOH$ or 50% $KOH/NaOH$

ⓐ Self cannizzaro rxn :-

$$2H-\overset{O}{\overset{\|}{C}}-H \xrightarrow[KOH]{conc.} H-\overset{O}{\overset{\|}{C}}-OK + H-CH_2-OH$$

* OP Point :-

① $H^{\ominus}$ transfer is the R.D.S

② example of Disproportionation Rxn

③ $H^{\ominus}$ & $H^{\oplus}$ Both type of transfer.

D.M

$-\overset{O}{\overset{\|}{C}}-H \xrightarrow[{[O]}]{oxi} -\overset{O}{\overset{\|}{C}}-O^-K^+$

$-\overset{O}{\overset{\|}{C}}-H \xrightarrow{Red} -CH_2-OH$

Alco $\underset{Red}{\overset{oxi}{\rightleftarrows}}$ Ald $\underset{Red}{\overset{oxi}{\rightleftarrows}}$ Acid

ⓑ Cross cannizzaro rxn :-

oxi → Jo NAR fast dega

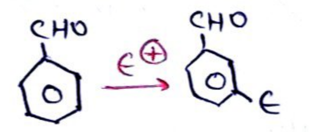
Red → 2nd wale ka Red^n

$$H-\overset{O}{\overset{\|}{C}}-H + Ph-\overset{O}{\overset{\|}{C}}-H \xrightarrow[KOH]{Scf.} HCOOK + Ph-CH_2OH$$

oxi, Red^n

↳ fastest NAR → [HCHO > PhCHO]

④ E.S.R



CHO → −M
Meta director

MOP of Carboxylic Acid

① from oxid^n of Alc., Ald. & ketones

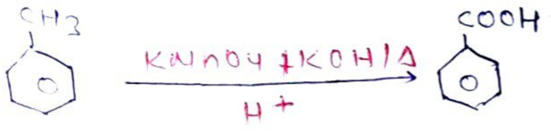
1° Alcohol $\xrightarrow{S.O.A}$ COOH

Aldehyde $\xrightarrow[T.R\ \&\ F.R\ \&\ B.R]{SOA}$ COOH

Ketone $\xrightarrow[rule]{Popoff's}$ COOH

② from Oxidⁿ of Alkylated benzene

↳ At least one Benzylic 'H'



③ from Hydrolysis of Acid derivatives

ⓐ Nitrile

(i) Partial hydrolysis

$$R-C\equiv N \xrightarrow[\Delta]{H_3O^+} R-\overset{O}{\overset{||}{C}}-NH_2$$

CN Hatao CONH₂ lagao

(ii) Complete hydrolysis

$$R-CN \xrightarrow[\Delta]{H_3O^+} R-COOH$$

CN Hatao COOH lagao

ⓑ Acid chloride

$$R-\overset{O}{\overset{||}{C}}-Cl \xrightarrow{H_3O^+} R-COOH$$

Cl hatao OH lagao

ⓒ Anhydride

$$R-\overset{O}{\overset{||}{C}}-OCOR' \xrightarrow{H_3O^+} RCOOH + R'COOH$$

ⓓ Ester

$$R-\overset{O}{\overset{||}{C}}-OR' \xrightarrow{H_3O^+} RCOOH + R'OH$$

OR' hatao OH lagao

ⓔ Amides

$$R-\overset{O}{\overset{||}{C}}-NH_2 \xrightarrow{H_3O^+} R-COOH$$

NH₂ hatao OH lagao

Saponification → Alkaline hydrolysis of Ester

$$R-\overset{O}{\overset{||}{C}}-OR' \xrightarrow[H_2O]{NaOH/} RCOONa + R'OH$$

$\downarrow H^+$

$R-COOH$

④ from Grignard Reagent

$$RMgX \xrightarrow[H^+]{CO_2} R-COOH$$

Properties of Carboxylic Acid

① Acidic strength

$$AS \propto -M, -H, -I \propto Ka \propto \frac{1}{pKa} \propto [H^+] \propto \frac{1}{pH}$$

$$R-COOH \longrightarrow R-COO^{\ominus} + H^+$$

② Rxn with $PCl_3, PCl_5, SOCl_2$

$$R-\overset{O}{\overset{||}{C}}-OH \xrightarrow[\text{or } SOCl_2]{PCl_3 \text{ or } PCl_5} R-\overset{O}{\overset{||}{C}}-Cl$$

③ Anhydride formation

$$2R-\overset{O}{\overset{||}{C}}-OH \xrightarrow[\text{or } P_2O_5 \text{ or } P_4O_{10}]{\text{Conc. } H_2SO_4/\Delta} R-\overset{O}{\overset{||}{C}}-O-\overset{O}{\overset{||}{C}}-R$$

④ Esterification

$$R-\overset{O}{\overset{||}{C}}-OH + HOR' \overset{H^+}{\rightleftharpoons} R-\overset{O}{\overset{||}{C}}-OR' + H_2O$$

⑤ Rxn with Ammonia

$$R-\overset{O}{\overset{||}{C}}-OH + NH_3 \underset{-H_2O}{\overset{\Delta}{\rightleftharpoons}} R-\overset{O}{\overset{||}{C}}-NH_2$$

⑥ Hell-Volhard-Zelinsky Rxn (HVZ)

↳ Acid having at least 1 α-H.

$$R-\underset{H}{\underset{|}{CH}}-COOH \xrightarrow[H_2O]{\text{Red } P + X_2} R-\underset{X}{\underset{|}{CH}}-COOH$$

α-halo Acid

★ OP Point :-

$$R-CH\begin{matrix}COOH\\X\end{matrix} \xrightarrow[NH_3]{(\text{excess})} R-CH\begin{matrix}COONH_4\\NH_2\end{matrix}$$

$\downarrow H_2O$

$\xrightarrow{NH_3}$ α-Amino Acid $R-\overset{\alpha}{CH}\begin{matrix}COOH\\NH_2\end{matrix}$

⑦ Kolbe's Electrolysis

$$R-COONa \xrightarrow{\text{Electro}} R-R$$

⑧ Decarboxylation

$$R-COOH \xrightarrow{NaOH/CaO/\Delta} R-H$$

$$R-COONa \xrightarrow{\text{"}} R-H$$

⑨ ESR

COOH → -M Meta director

⑩ Reduction

$$R-COOH \xrightarrow[\text{or}]{LiAlH_4} R-CH_2OH$$

B_2H_6 → only acid ko reduce krta h.

★ OP Point :-

↳ Reduction rxn of HI / Red P (S.R.A)

$$R-\overset{O}{\overset{||}{C}}-OH \longrightarrow R-CH_3$$

$$R-\overset{O}{\overset{||}{C}}-H \longrightarrow R-CH_3$$

$$R-\overset{O}{\overset{||}{C}}-R' \longrightarrow R-CH_2-R'$$

$$R-OH \longrightarrow R-H$$